

Generalist Models in Medical Image Segmentation: A Survey and Performance Comparison with Task-Specific Approaches

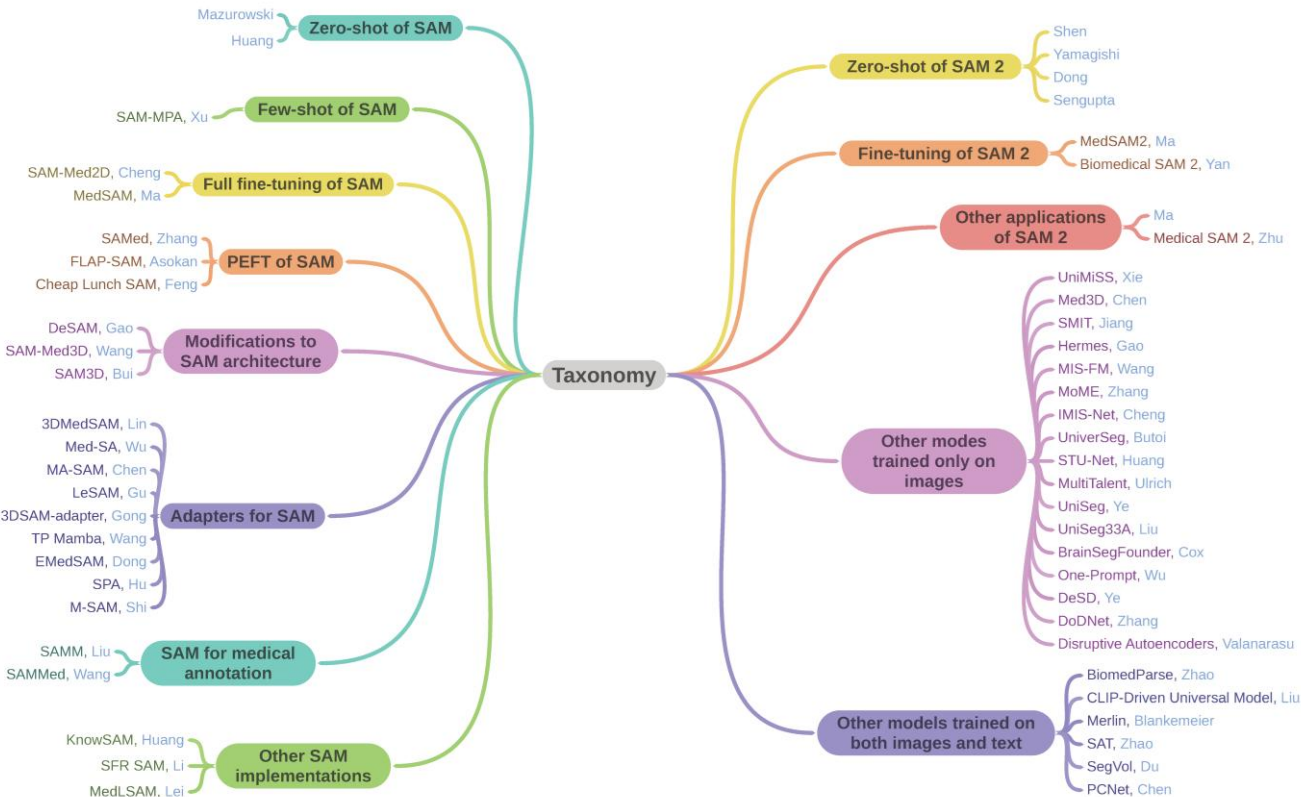
Fundamentals
on generalist
models

Taxonomy

Performance
comparison with SOTA
task-specific models

Challenges

Future
directions



Contributions

- ✓ Comprehensive architectural dissection of generalist models;
- ✓ Assessment of performance progress from primary research to best-in-literature for both generalist and SOTA task-specific models;
- ✓ Dataset-wise and organ-wise performance comparison between generalist and SOTA task-specific models;
- ✓ Challenges on compliance with regulatory frameworks, privacy and security laws, budget, and trustworthy AI;
- ✓ Future directions on synthetic data, early fusion, lessons learnt from generalist models in natural language processing, agentic AI and physical AI, and clinical translation.